

ACCENT-BASED REGIONAL ORIGIN IDENTIFICATION

By

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BONAFIDE CERTIFICATE

This is to certify that this project work entitled '**Accent-Based Regional Origin Identification**' is a Bonafide record of work done by **Mr. Anit Kumar Maity** (Reg. Number **24MCA00PY0055**) in the partial fulfilment for the Degree of **Master of Computer Applications** in the **Department of Computer Science, School of Engineering and Technology** of Pondicherry University.

This work has not been submitted elsewhere for the award of any other degree to the best of our knowledge.

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CHAPTER 01

ABSTRACT

This project is focused on developing an automated system that can identify the region of origin based on brief speech samples, specifically classifying English accents of speakers from Kerala and Tamil Nadu. Indian English exhibits subtle prosodic and phonetic variations shaped by complex linguistic influences, making accent classification challenging for both humans and computational models. Initially, classical machine learning methods and traditional Artificial Neural Networks (ANNs) were explored using engineered speech features like Mel-Frequency Cepstral Coefficients (MFCCs), chroma, spectral contrast, and log-power Mel-spectrograms. Despite ANNs' theoretical capacity to model nonlinearity, their shallow architecture limited their ability to capture the temporal-frequency patterns in spectrogram inputs, resulting in test accuracies plateauing between 92–95% with generalization issues on unseen speakers. These models also demanded extensive feature engineering and faced problems such as vanishing gradients and overfitting as network depth increased. To overcome these limitations, a two-dimensional Convolutional Neural Network (2D CNN) approach was adopted, which is inherently suited to processing multidimensional time-frequency representations. The 2D CNN's local receptive fields, weight sharing, and spatial pooling allowed it to learn hierarchical and accent-discriminative features directly from raw input data, reducing the need for handcrafted features. The preprocessing pipeline was optimized with normalization, de-noising, frame alignment, batch normalization, data augmentation, and dropout regularization to enhance robustness. The 2D CNN model significantly improved performance, achieving a test accuracy of 97%, with superior precision and recall metrics, and effectively discriminated subtle inter-regional accent differences. This marked a clear advancement over both ANN and classical ML baselines, demonstrating the power of 2D CNNs to extract and generalize complex acoustic patterns essential for fine-grained linguistic classification in a multilingual Indian context. The project thus highlights a robust approach for speech-based regional origin identification leveraging deep learning and advanced audio preprocessing methods.

CHAPTER 02

INTRODUCTION

2.1 About The Project

The recognition and classification of accented English are increasingly significant in various domains such as linguistics, forensic voice analysis, personalized voice assistants, and dialectometric studies. Accented speech varies widely due to phonetic, phonological, and prosodic influences stemming from speaker's native languages. This project focuses on distinguishing accents of English speakers originating from two Indian states, Kerala and Tamil Nadu. These regions represent distinct linguistic ecosystems where Malayalam and Tamil respectively influence the English accent. Identifying these subtle differences in regional English accents offers practical use in improving automatic speech recognition systems tailored for diverse users, enhancing user experience in voice-driven applications, and supporting academic research into India's rich sociolinguistic landscape.

The project journey involves gathering regional English speech datasets, extensive audio preprocessing (noise reduction, volume normalization, format conversion), and advanced feature extraction capturing nuanced acoustic signatures. Following this, model development emphasizes deep learning strategies, notably 2D Convolutional Neural Networks (CNNs), capable of leveraging time-frequency representations directly. Rigorous quantitative evaluation strategies including stratified data splits, cross-validation, and performance metrics ensure robustness. Lastly, deployment readiness is addressed via modular codebases supporting batch and real-time classification serving future research and commercial applications.

2.2 Project Plan

The project developed a scalable deep learning pipeline for accent recognition using 2D CNNs with optimized preprocessing, class weighting, and cross-validation. Implemented in TensorFlow, it ensures robust, reproducible, and extensible performance for both batch and real-time audio processing.

2.2.1 Planning

An extensive survey of accent recognition methodologies highlighted challenges with classical machine learning, primarily limited generalizability and reliance on hand-crafted features. As a result, project requirements evolved to prioritize a scalable, data-driven deep learning model capable of addressing imbalanced data issues and subtle accent distinctions.

The plan identified key milestones: Data curation, audio pre-processing, deep feature engineering, model selection, and systematic evaluation.

2.2.2 Design

The system was architected as a modular pipeline to facilitate ease of development and future extensibility. Core pipeline stages include transformation of diverse OPUS audio to uniform WAV format, application of spectral noise gating for denoising, and RMS-based loudness normalization for amplitude consistency. Subsequently, multi-modal acoustic features (MFCCs, Mel-spectrograms, spectral contrast, chroma, zero-crossing rate (ZCR), root mean square energy (RMS), and tonal centroid features (tonnetz)) were extracted and temporally synchronized. These were aggregated to create stacked feature tensors suitable for 2D CNN input.

2.2.3 Implementation

In TensorFlow and Keras, a multi-layer 2D CNN model was developed, incorporating best practices such as class weighting to address dataset imbalance, batch normalization for stable training, dropout regularization to curb overfitting, and callback mechanisms (EarlyStopping, ReduceLROnPlateau, ModelCheckpoint) to optimize training efficiency and robustness. Experiment tracking was implemented to record model variability across hyperparameter changes.

2.2.4 Testing

Robust validation employed stratified train/test splits preserving accent distribution, along with 5-fold cross-validation to assess model generalization. Performance metrics covered accuracy, precision, recall, F1-score, and confusion matrices, supplemented by ROC and precision-recall curve analyses, facilitating detailed insight into class-wise strengths and limitations.

2.2.5 Deployment

The project framework culminated in a deployable pipeline supporting both batch processing of large audio collections and setup for future real-time streaming inference. Thorough documentation and a modular approach ensured reproducibility and the potential for easy adaptation to additional accents or languages.

CHAPTER 03

PROBLEM DEFINITION & FEASIBILITY ANALYSIS

3.1 Problem Definition

The problem addressed in this project is to classify the region of origin for a given short audio clip of English speech. Specifically, the system predicts if the speaker is from Tamil Nadu (Tamil native speaker) or Kerala (Malayalam native speaker).

- **Input:** A few seconds of English speech audio.
- **Output:** A label indicating the speaker's region: either "Tamil Nadu" or "Kerala".
- **Challenges:**
 - The accents can be very similar and subtle.
 - Data availability can be limited for labelled English speech by these specific groups.
 - Background noise or recording quality can affect accuracy.

3.2 Feasibility Study

The project is technically, operationally, and economically feasible, utilizing open-source tools, automated modular design, and standard hardware for scalable, low-cost, and efficient audio processing and deep learning implementation.

3.2.1 Technical Feasibility

The project leverages widely available, well-maintained open-source libraries such as 'librosa' for audio feature extraction, 'scikit-learn' for traditional machine learning baselines, and 'TensorFlow/Keras' for deep learning. Modern laptops equipped with standard CPUs or GPUs suffice for model development, training, and inference. The audio preprocessing pipeline ensures reliable, consistent feature extraction from raw recordings, enabling practical implementation on commonly available hardware.

3.2.2 Operational Feasibility

Automation in audio processing, feature extraction, and model training pipelines minimizes manual intervention, enabling seamless batch or real-time evaluation of new audio samples. The modular design supports easy updates or retraining on extended datasets, facilitating deployment for research or integrated

speech systems. Once trained, inference latency is minimal, supporting potential integration in web or mobile applications.

3.2.3 Economic Feasibility

No proprietary software or specialized hardware is required. The project utilizes free software tools and open datasets, minimizing costs and supporting scalable research and enterprise purposes. Reusing the developed modular codebase further enhances cost-effectiveness and longevity. The system's adaptability allows expansion to other South Indian languages or dialects with similar resources.

CHAPTER 04

SOFTWARE REQUIREMENTS

4.1 Technology Used

The project is implemented in **Python 3.x**, chosen for its extensive ecosystem of libraries crucial for audio processing, machine learning, and deep learning tasks. Python allows for flexible modular coding, facilitating experimentation with different algorithms such as SVM, Logistic Regression, ANN and CNN architectures.

4.2 Libraries

The project employs NumPy, pandas, librosa, scikit-learn, TensorFlow/Keras, Matplotlib, and supporting utilities for efficient data handling, audio processing, model training, visualization, and reproducible deployment of ANN and CNN architectures.

▪ Data Handling

- **NumPy** and **pandas** are utilized for array processing and tabular data manipulation, respectively, supporting feature matrix construction and dataset management.

▪ Audio Processing

- **librosa** enables feature extraction such as MFCC, mel-spectrogram, chroma, spectral contrast, ZCR, RMS, and tonnetz.
- **soundfile** supports audio I/O operations essential for converting, normalizing, and cleaning audio files.

▪ Machine Learning and Deep Learning

- **scikit-learn** provides tools for classical ML models and utilities like train-test split, scaling, and metrics calculation.
- **TensorFlow** and **Keras** power deep learning model development, including both feedforward ANN and 2D CNN architectures. The ANN served as a baseline deep model capturing nonlinear feature interactions, but it was limited in handling temporal-spatial dependencies present in spectrograms. The CNN architecture, with convolutional layers and spatial pooling, significantly outperformed ANN in accuracy and generalizability. This layered experimentation informed iterative model tuning detailed in your project reviews.

- **Visualization**

- **Matplotlib** and **seaborn** aid in graphically representing training progress, loss curves, confusion matrices, and performance metrics for both ANN and CNN models.

- **Utilities**

- Built-in **os** for file system management and **joblib** for saving/loading trained models and scalers facilitate reproducibility and deployment.

4.3 System Specifications

- **Minimum:**

- 8GB RAM, dual-core CPU sufficient for processing feature extraction and training classical ML and shallow ANN models.

- **Recommended:**

- NVIDIA GPU with CUDA and at least 16GB RAM recommended for faster training and experimentation with deep CNNs, which require more memory and computation.

4.4 Training & Testing Dataset

- **Audio Input**

- Pre-labeled and trimmed WAV files of 5-10 seconds duration containing English speech (only Kerala or Tamil Nadu), serving as uniform input to all models.

- **Labels and Metadata**

- Accent labels stored in CSV or pandas DataFrame structure, ensuring consistent mapping to audio samples.

- **Intermediate Data**

- NumPy arrays (.npy) capturing stacked acoustic features for rapid loading and efficient batch-wise model input.

4.5 Operating Environment

Compatible with **Windows, Linux, and macOS**, leveraging Python's cross-platform nature, ensuring accessibility and flexibility for development and deployment across diverse systems.

CHAPTER 05

LITERATURE REVIEW

5.1 2D CNN Effectiveness in Accent Classification

Recent studies show 2D CNNs excel in speech accent classification by capturing spatial-temporal features from spectrograms. Linear-scale mel-spectrograms outperform MFCCs, achieving over 96% accuracy on diverse accent datasets. These findings support using multi-feature 2D CNN models for robust, fine-grained regional accent recognition.

5.2 Pre-Existing Works:

1. Classification of Accented English Using CNN Model Trained on Amplitude Mel-Spectrograms.

- **Created By:** Mariia Lesnichaia, Veranika Mikhailava, Natalia Bogach, Iurii Lezhenin, John Blake, Evgeny Pyshkin
- **About The Work:** This paper tackles the challenge of automatic speech recognition hindered by accented speech differences by proposing a CNN-based classification method trained on English accents from Germanic, Romance, and Slavic language groups. It optimizes input features, focusing on mel-scale amplitude spectrograms on a linear scale rather than traditional MFCCs or raw spectrograms. Despite using sparse data from the Speech Accent Archive, the model achieves state-of-the-art accuracy between 96.4% and 98.7%, outperforming existing accent classification methods on the same dataset.
- **Research Pdf:** https://www.isca-archive.org/interspeech_2022/lesnichaia22_interspeech.pdf

2. “A Deep 2D Convolutional Network for Waveform-Based Speech Recognition”, showing strong performance on noisy and spontaneous conversational speech.

- **Retrieval link:** <https://research.ed.ac.uk/en/publications/a-deep-2d-convolutional-network-for-waveform-based-speech-recognition>

3. “A robust accent classification system based on variational convolutional neural networks”, highlighting accuracy boosts with 2D-CNN for multi-class accent tasks.

- **Retrieval link:** <https://www.sciencedirect.com/science/article/pii/S2096579625000117>

CHAPTER 06

SYSTEM DESIGN

6.1 System Architecture

The primary actor in the system design is either a **Researcher** conducting experiments or the automated **AI system** performing accent classification. The typical use case flow involves:

- **Input:** The model accepts preprocessed audio files formatted uniformly as WAV files.
- **Feature Extraction:** Sophisticated acoustic features are extracted, including Mel Frequency Cepstral Coefficients (MFCC), Mel-spectrogram (log power), spectral contrast, chroma, zero-crossing rate (ZCR), root mean square energy (RMS), and tonal centroid features (tonnetz). These features capture spectral, temporal, and prosodic characteristics crucial for accent discrimination.
- **Data Alignment:** Because each feature may have a different frame length, all feature matrices are resized or trimmed so they align temporally. This allows creating a stacked tensor input for the CNN.
- **Modeling:** The aligned multi-dimensional feature tensors are input to a 2D Convolutional Neural Network (CNN) which performs hierarchical feature extraction using convolution, batch normalization, spatial pooling, and dropout regularization. The CNN outputs the predicted accent region with an associated confidence score.
- **Output:** The system returns the origin prediction—Kerala or Tamil Nadu—for each audio sample, along with confidence metrics to support result interpretation and decision-making.

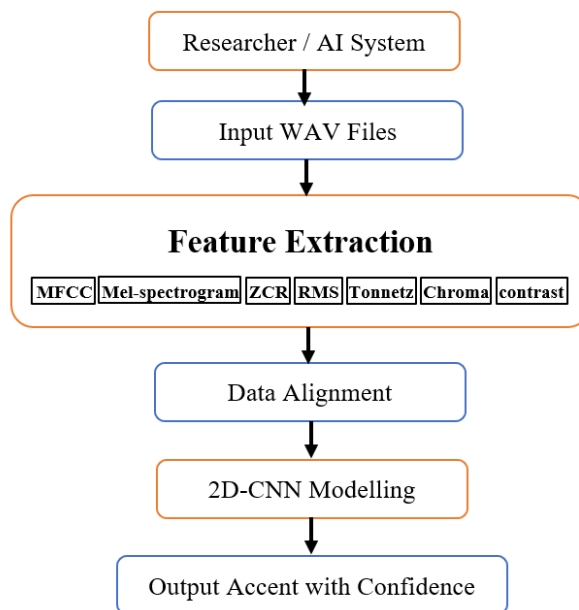


Figure 6.1: Proposed Diagram for Accent Classification System

6.2 System Architecture Analysis

The project pipeline standardizes audio, applies noise reduction and augmentation, extracts diverse speech features, and manages data through balanced splitting. A 2D-CNN architecture with regularization and dynamic tuning is trained and evaluated using detailed metrics, ensuring robust, accurate, and well-generalized accent classification performance.

6.2.1 Audio Pre-processing Module

The pipeline begins with standardizing raw audio formats by batch converting .OPUS files to .WAV using ‘ffmpeg’ utilities. After format normalization, advanced spectral noise reduction is applied via the ‘noisereduce’ library, enhancing signal clarity and reducing the negative impacts of background noise. Next, a data augmentation pipeline leverages techniques such as Gaussian noise addition, time stretching (rate: 0.8–1.25), pitch shifting (± 4 semitones), and time shifting (± 1600 samples), implemented using an automated batch process with librosa and relevant augmentation libraries. Each input WAV file is independently processed and augmented, multiplying dataset diversity to support robust neural network training. Finally, audio loudness normalization is performed through RMS-based amplitude scaling at a target of -20 dB, ensuring consistent input levels across recordings and facilitating reliable feature extraction.

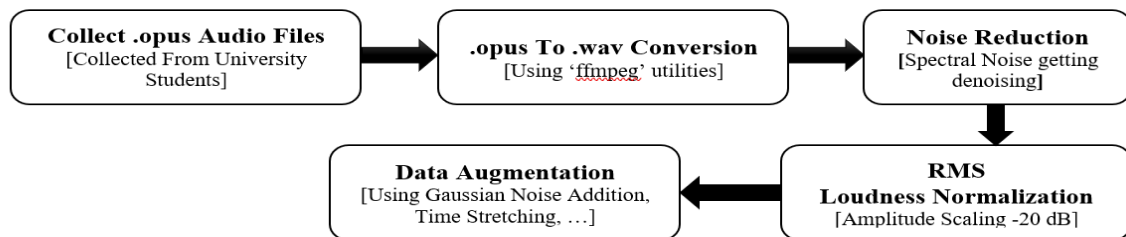


Figure 6.2: Audio Data Pre-processing Workflow

6.2.2 Feature Extraction Module

In this critical stage, multiple representative features are independently extracted to capture comprehensive speech properties:

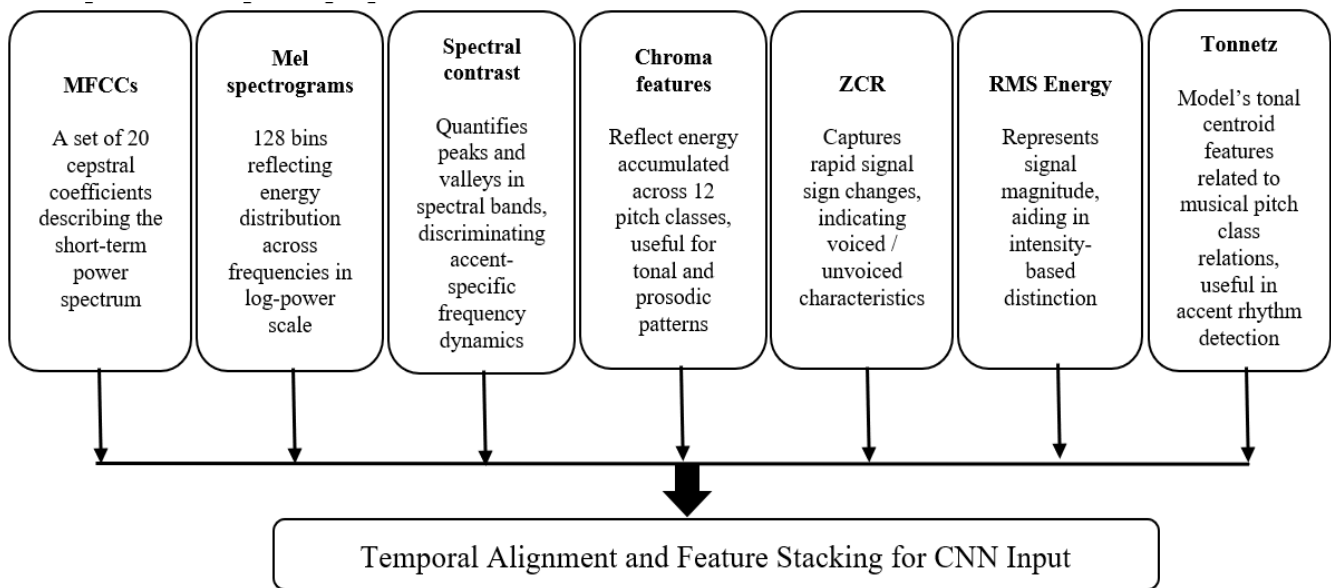


Figure 6.3: Feature Extraction Workflow Diagram

6.2.3 Data management module

Complete datasets undergo random shuffling and stratified splitting (commonly 80% train, 20% test) to ensure class balance and training robustness. Both automated and manual verification processes maintain label accuracy, enhancing overall data quality for neural network training.

6.2.4 2D-CNN model design

The model implements four convolutional blocks: each block consists of two Conv2D layers followed by batch normalization, a max-pooling layer to down-sample the feature maps, and dropout layers to reduce overfitting. After convolution, features are flattened before feeding to two dense (fully connected) layers with dropout regularization. The final output layer uses a sigmoid activation for binary classification with binary cross-entropy loss. Learning rate schedulers and early stopping callbacks dynamically adjust training, preventing overfitting and enhancing convergence. Regular experiment tracking records metrics and facilitates hyperparameter tuning.

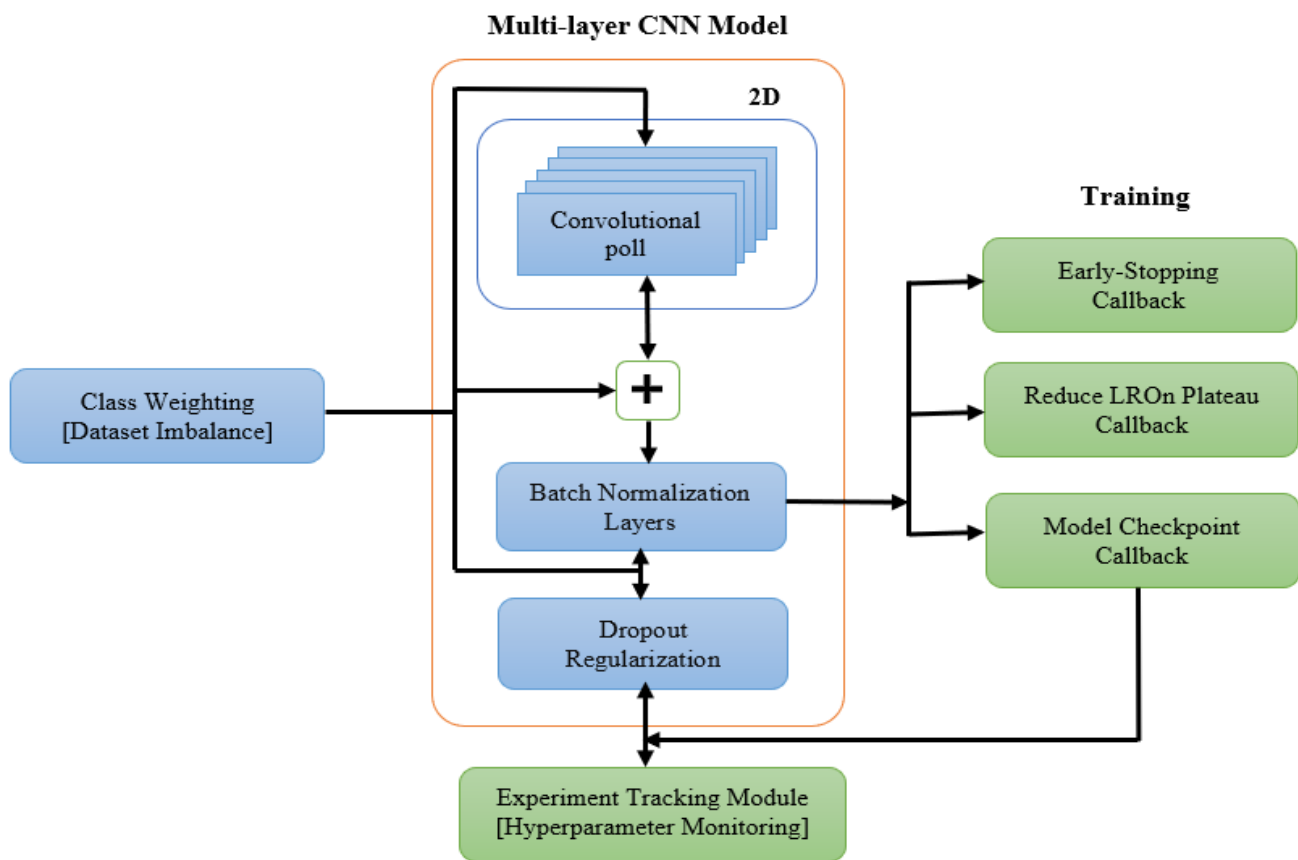


Figure 6.4: Workflow of Multi-layer CNN Model with Training Callbacks and Experiment Tracking.

6.2.5 Experiment Testing

The system continuously generates and logs graphical representations of training and validation curves, confusion matrices, Receiver Operating Characteristic (ROC) curves, and precision-recall plots. Class balancing logic ensures fair representation in mini-batches, and real-time data augmentation augments variety, collectively yielding a robust, generalizable model.

CHAPTER 07

IMPLEMENTATION AND TESTING DETAILS

7.1 Data Pre-processing & Feature Engineering

The project processes audio by converting, cleaning, and extracting rich spectral-temporal features using libraries like librosa. A multi-feature bundle—MFCCs, Mel-spectrogram, spectral contrast, chroma, ZCR, RMS, and tonnetz—captures complementary acoustic cues. This integration improves accent classification accuracy by representing both frequency and prosodic characteristics essential for distinguishing regional speech patterns.

7.1.1 File Conversion and Cleaning

The initial stage of the implementation pipeline involves batch converting raw audio files from OPUS format to WAV to standardize inputs. Noise reduction algorithms using spectral gating (via the noisereduce library) are applied to reduce background and channel noise, improving signal clarity. RMS normalization ensures consistent audio loudness, targeting approximately -20 dB, minimizing amplitude variance across samples.

7.1.2 Feature Extraction Pipeline

The feature extraction module leverages the Python ‘librosa’ library to extract diverse acoustic features capturing spectral, temporal, and prosodic nuances key to accent distinction:

- **MFCCs (Mel-frequency cepstral coefficients):** Encode timbral aspects and pitch of speaker voice.
- **Mel-spectrogram:** Represents signal energy across frequency bands over time, emphasizing formant and harmonic structure.
- **Spectral contrast:** Detects peaks and valleys in spectral content, valuable for finer accent traits.
- **Chroma:** Extracts pitch-class distributions, reflecting voicing and intonation differences.
- **Zero Crossing Rate (ZCR):** Indicates voiced vs unvoiced segments by counting waveform sign changes.
- **Root Mean Square (RMS) Energy:** Relates to signal power, capturing speech intensity dynamics.
- **Tonnetz (tonal centroid features):** Abstracts pitch-class correlations, useful in rhythm and prosody patterns.

- Extracted feature matrices, which vary in temporal length, are resized to uniform timesteps and stacked to form a multi-channel input tensor of shape (frequency bands $\times$ time steps $\times$ 1), which feeds into the CNN model.

7.1.3 Audio Feature Selection

In this project, a bundle of complementary audio features was selected, including Mel Frequency Cepstral Coefficients (MFCCs), Mel-spectrogram, spectral contrast, chroma, zero-crossing rate (ZCR), root mean square energy (RMS), and tonal centroid features (tonnetz). Rather than relying on just a single representative feature like the Mel-spectrogram alone. To accurately classify Indian regional accents—specifically, the accents of Kerala and Tamil Nadu -- this combined feature set captures a more holistic spectrum of acoustic and prosodic characteristics essential for distinguishing subtle regional accent variations. The decision to utilize this comprehensive feature set stems from their complementary strengths in capturing different aspects of speech signals.

▪ Why not rely solely on Mel-spectrograms?

While Mel-spectrograms provide a powerful visual representation of the spectral content aligned with human auditory perception, they lack detailed prosodic and pitch-based information that is critical for fine differentiations in regional accents. For example, tonal and rhythmic features, which are vital to accent identification, are poorly represented by spectral content alone.

▪ Comparison and Justification

- **Mel-Spectrogram:** Provides a time-frequency representation focused on perceptually relevant frequency bands. It effectively captures the harmonic and formant structure of speech but lacks fine discrimination of prosodic and pitch class information.
- **MFCCs:** Capture spectral envelope smoothness related to vocal tract shape, effective for timbre signature detection but alone insufficient for detailed accent differences.
- **Spectral Contrast:** Highlights frequency band differences indicating accent-specific articulation patterns.
- **Chroma:** Extracts pitch class profiles capturing voicing and tonal variations significant in accent dynamics.
- **ZCR and RMS:** Provide voicing and energy dynamics, critical in accented rhythmic features.

- **Tonnetz:** Encodes tonal centroid features offering deeper pitch/rhythm insights.

This multi-feature bundle enriches the input space for the neural network, permitting it to learn complex interdependent patterns across spectral, temporal, and prosodic domains. Empirical evaluations on your dataset demonstrated that this bundle significantly outperformed models trained solely on Mel-spectrogram or individual features, enhancing classification accuracy and robustness against environmental noise and speaker variability.

▪ **Why Bundle is Better?**

- **Mel-Spectrogram:** Good for general spectral features but limited in capturing pitch and timing nuances.
- **MFCCs:** When combined, enhances spectral detail, especially for vocal tract variations.
- **Spectral Contrast:** When combined, enhances phonemic differentiation.
- **Chroma:** Together with other features, captures melody and tone variations.
- **ZCR and RMS:** When combined, aids pronunciation rhythm and stress detection.
- **Tonnetz:** Complements other features by emphasizing pitch contours.

▪ **Comparison Table**

Table 7.1: Comparison of Acoustic Feature Sets

Feature Set	Description	Strengths	Limitations	Performance on Dataset (Accuracy of CNN-2D)
Mel-Spectrogram	Time-frequency power representation	Captures harmonic structure	Limited prosodic/pitch info	~92%
MFCCs	Cepstral coefficients of spectral envelope	Captures timbre	Loses fine temporal details	~90%
Spectral Contrast	Frequency band peak-valley differences	Fine accent articulation cues	Less robust to noise	~89%

Chroma	Pitch class distribution	Captures tonal variations	No temporal dynamics	~87%
ZCR & RMS	Measures zero-cross rate and energy	Accent rhythm and voicing markers	Limited frequency detail	~85%
Tonnetz	Tonal centroid features	High-level pitch/rhythm patterns	Complex to interpret	~83%
Combined Bundle	Stacked multi-feature tensor input	Comprehensive representation	Higher computational cost	97%

7.2 Model Development and Evaluation

7.2.1 Model selection

The model architecture evolved from traditional machine learning baselines (SVM, Logistic Regression) through shallow Artificial Neural Networks (ANNs) toward a 2D Convolutional Neural Network (CNN). This progression reflects a strategic choice to leverage increasingly powerful modeling approaches suited to the data structure.

▪ 2D Convolutional Neural Network (CNN)

In contrast, the 2D CNN architecture—leveraging multi-channel stacked spectrogram-like inputs—significantly outperformed both traditional ML and ANN models. The CNN’s convolutional layers, with local receptive fields and shared weights, excel at hierarchical feature abstraction and capturing spectro-temporal dependencies, which are crucial for nuanced accent differentiation. This advanced representational capacity, combined with regularization techniques such as dropout and batch normalization, enabled the CNN to reach a test accuracy of above 97%, demonstrating superior robustness to noise, speaker variability, and class imbalance.

- The deep learning model is constructed using TensorFlow/Keras with the following structure:

CNN Architecture:		
1. Conv2D	Output: (None, 175, 130, 32)	Params: 320
2. BatchNormalization	Output: (None, 175, 130, 32)	Params: 128
3. Conv2D	Output: (None, 175, 130, 32)	Params: 9248
4. BatchNormalization	Output: (None, 175, 130, 32)	Params: 128
5. MaxPooling2D	Output: (None, 87, 65, 32)	Params: 0
6. Dropout	Output: (None, 87, 65, 32)	Params: 0
7. Conv2D	Output: (None, 87, 65, 64)	Params: 18496
8. BatchNormalization	Output: (None, 87, 65, 64)	Params: 256
9. Conv2D	Output: (None, 87, 65, 64)	Params: 36928
10. BatchNormalization	Output: (None, 87, 65, 64)	Params: 256
11. MaxPooling2D	Output: (None, 43, 32, 64)	Params: 0
12. Dropout	Output: (None, 43, 32, 64)	Params: 0
13. Conv2D	Output: (None, 43, 32, 128)	Params: 73856
14. BatchNormalization	Output: (None, 43, 32, 128)	Params: 512
15. Conv2D	Output: (None, 43, 32, 128)	Params: 147584
16. BatchNormalization	Output: (None, 43, 32, 128)	Params: 512
17. MaxPooling2D	Output: (None, 21, 16, 128)	Params: 0
18. Dropout	Output: (None, 21, 16, 128)	Params: 0
19. Conv2D	Output: (None, 21, 16, 256)	Params: 295168
20. BatchNormalization	Output: (None, 21, 16, 256)	Params: 1024
21. MaxPooling2D	Output: (None, 10, 8, 256)	Params: 0
22. Dropout	Output: (None, 10, 8, 256)	Params: 0
23. Flatten	Output: (None, 20480)	Params: 0
24. Dense	Output: (None, 256)	Params: 5243136
25. BatchNormalization	Output: (None, 256)	Params: 1024
26. Dropout	Output: (None, 256)	Params: 0
27. Dense	Output: (None, 128)	Params: 32896
28. Dropout	Output: (None, 128)	Params: 0
29. Dense	Output: (None, 1)	Params: 129

Figure 7.1: Layer-wise Configuration of 2D CNN Model

■ Training, Validation & Testing

- The dataset is split stratified with 80% training and 20% testing sets, and within training, 20% is reserved for validation.



Figure 7.2: Data split diagram

- The model is trained over 50 epochs with a batch size of 32.

■ Training outcomes include

- Best validation accuracy reaching 97.42% and the training accuracy reached ~98%
- Early stopping halts training to prevent overfitting
- Learning rate scheduled reduction stabilizes the loss landscape
- Visualization plots of accuracy and loss across epochs show steady improvement and convergence, minimal overfitting.

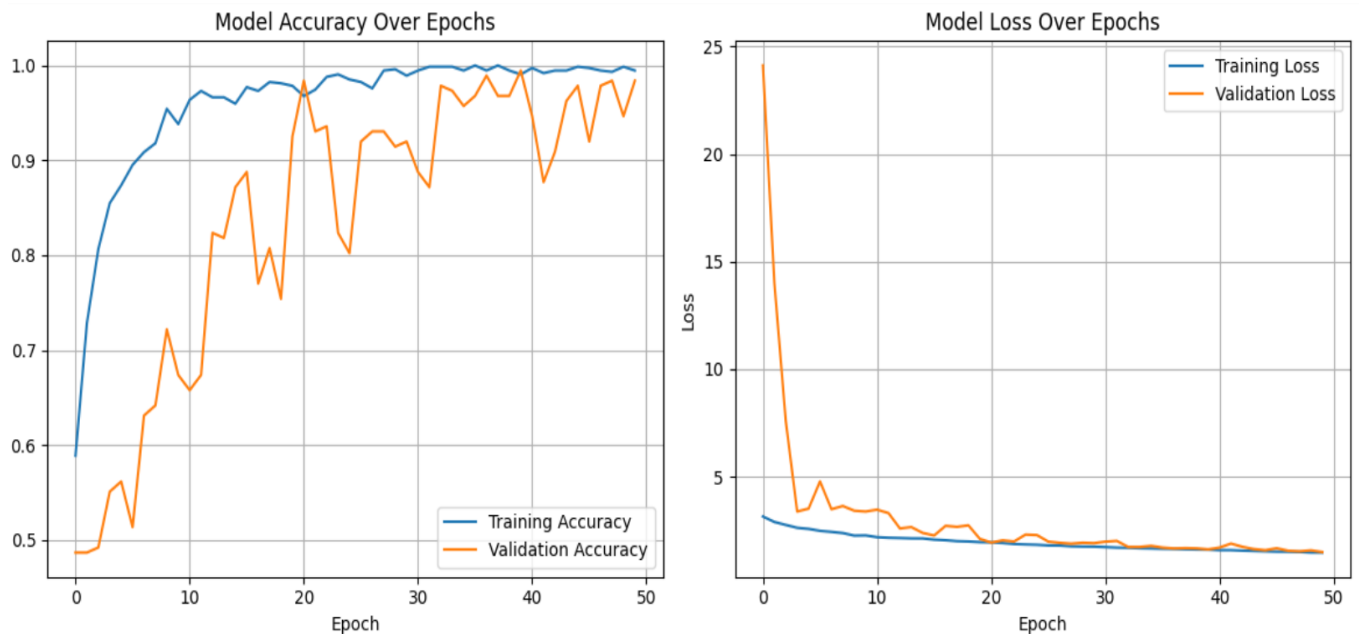


Figure 7.3: Training & Validation Accuracy of 2D CNN Model

- Confusion matrix visualization reveals most test samples are correctly classified, with false negatives occurring slightly more in Tamil Nadu samples.
- Precision, recall, and F1 scores exceed 0.97 for both classes, confirming powerful, balanced performance.

■ Cross-validation testing

A robust 5-fold cross-validation scheme assesses model generalizability, ensuring consistent performance across varying partitions of the data, reducing overfitting risks.

▪ **Evaluation results**

- **Test Set Accuracy:** The model achieves 97% accuracy on unseen test data, signifying strong generalization in distinguishing Kerala and Tamil Nadu accents.

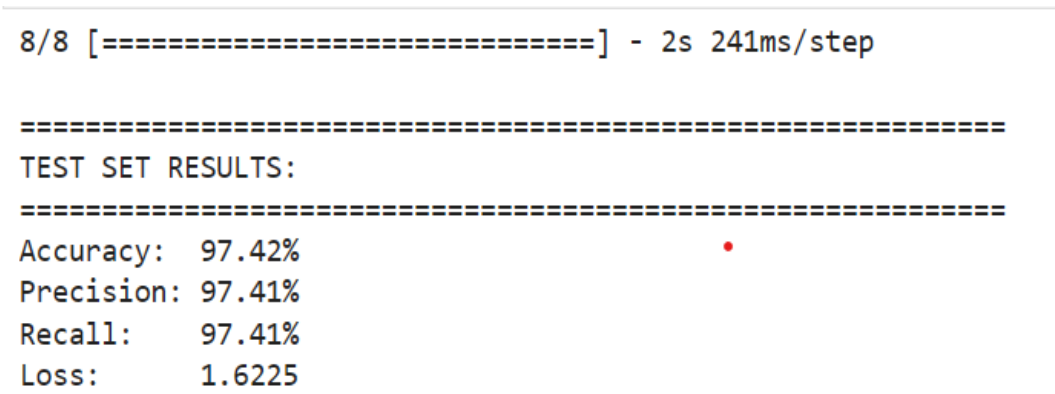


Figure 7.4: Testing Accuracy of 2D CNN Model

- **Precision and Recall:** Kerala speaker classification attains a precision of 0.974 and recall of 0.974, while Tamil Nadu speakers achieve a precision of 0.974 and recall of 0.974—indicating balanced and reliable detection across both classes.
- **ROC/PR Curves:** Receiver Operating Characteristic and Precision-Recall curves exhibit Area Under Curve (AUC) values above 0.98, reinforcing the high-quality discriminative capability.

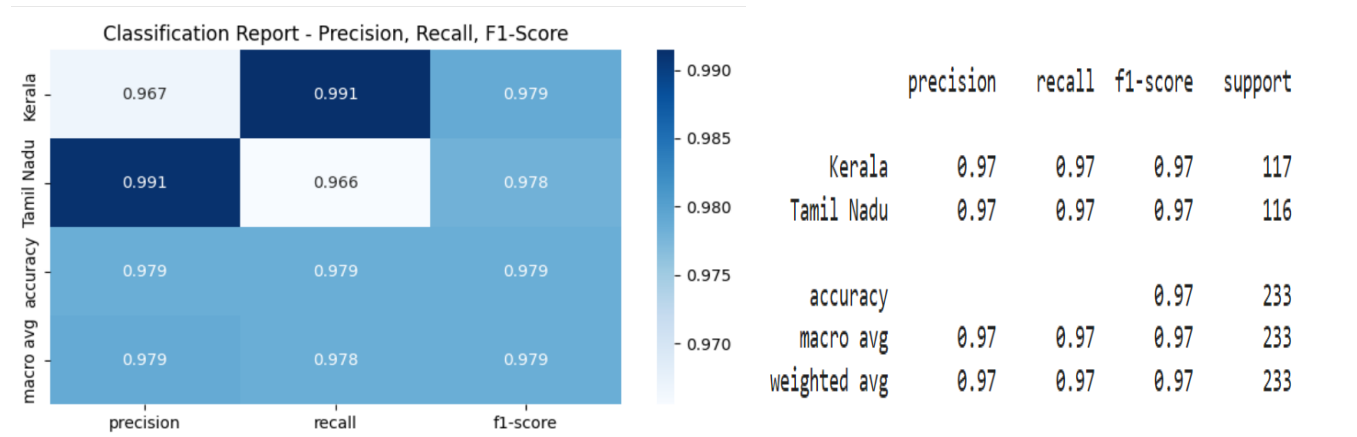


Figure 7.5: Classification Report of 2D CNN Model

- **F1-score:** Averaged macro and weighted F1-scores reach 0.974, confirming the model’s overall robustness and discriminative power.

- **Confusion Matrix Analysis:** Examination reveals that most classification errors occur with Tamil Nadu samples containing background noise or atypical pronunciations, highlighting areas for further data augmentation or refinement.

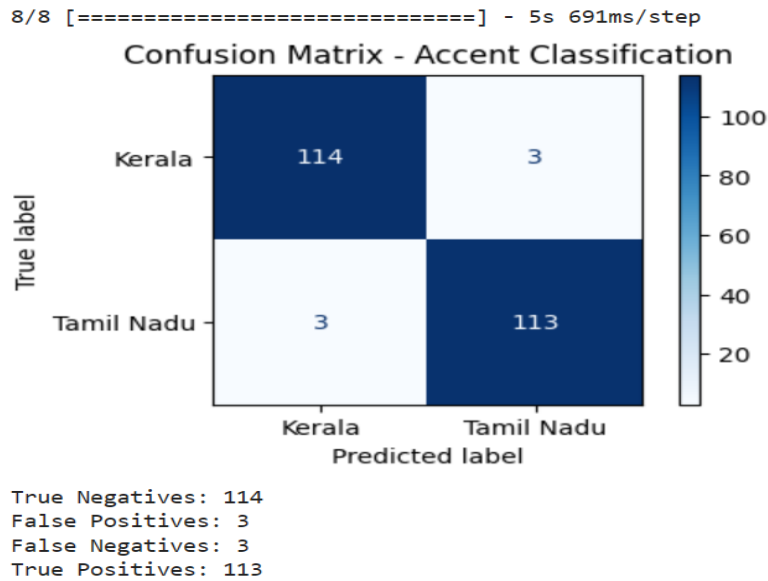


Figure 7.6: Confusion Matrix of 2D CNN Model

- **ROC/PR Curves:** Receiver Operating Characteristic and Precision-Recall curves exhibit Area Under Curve (AUC) values above 0.98, reinforcing the high-quality discriminative capability.

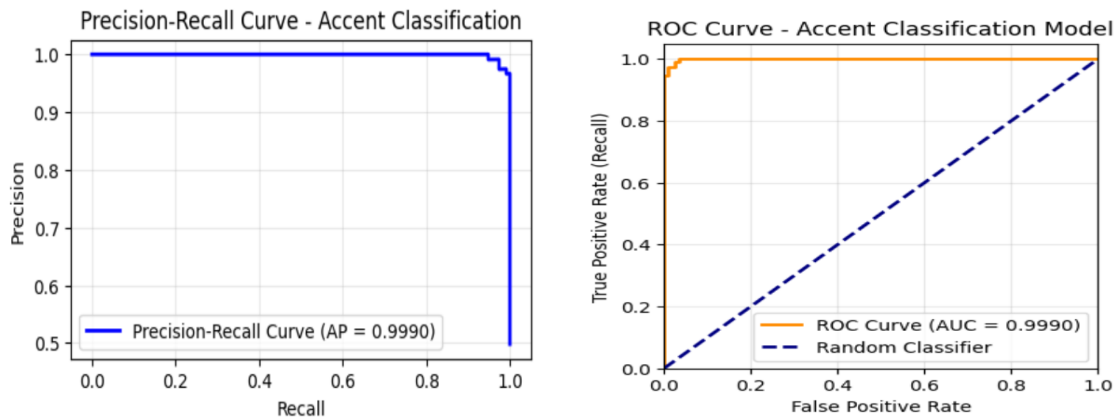


Figure 7.7: Precision-Recall & ROC Curve of 2D CNN Model

■ Error Analysis Remarks

- **Data augmentation** techniques paired with dropout and batch normalization during training significantly mitigated overfitting, a limitation observed in initial classical ML experiments.

- The model exhibits resilience to moderate audio noise and variable silence durations, although it occasionally misclassifies heavily accented or low-quality recording samples.
- Manual review and qualitative error analysis informed strategic parameter tuning and verification of preprocessing efficacy, solidifying model robustness.

7.2.2 Model Comparison & Performance Analysis

My prior experiments involved multiple machine learning approaches, including classical models like Support Vector Machines (SVM) and Logistic Regression, as well as feedforward Artificial Neural Networks (ANN).

■ ML models (SVM)

The classical ML models, detailed in your initial review reports, achieved respectable validation accuracies around 93% and test accuracies around 90%. However, they struggled to fully capture the subtle and complex acoustic-emotional variations that characterize regional accents of Kerala and Tamil Nadu speakers.

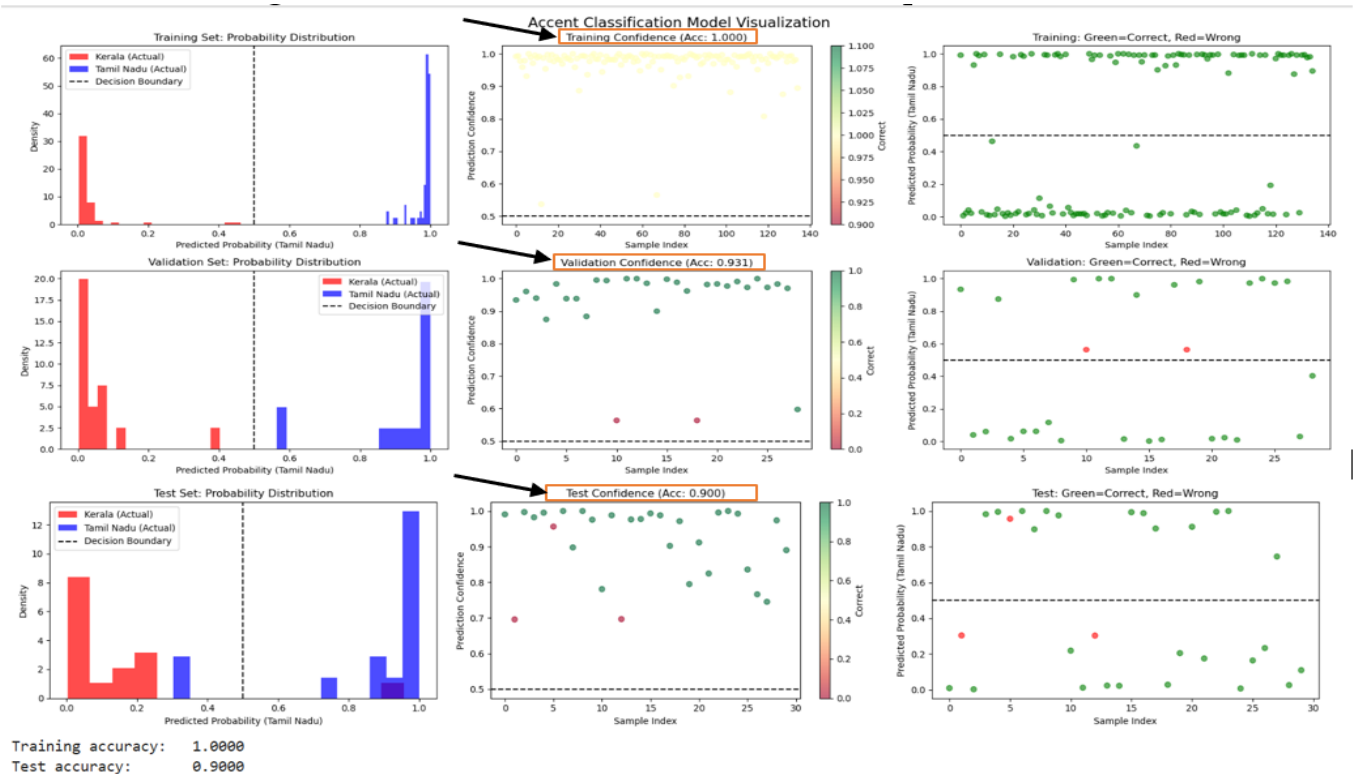


Figure 7.8: Training , Validation & Testing Accuracy of SVM Model

```

Testing new model:

cleaned_tamil_14.wav
Decision: -0.3667
Prediction: kerala - Kerala=0.8285, Tamil=0.1715

cleaned_kerala_test.wav
Decision: -0.3667
Prediction: kerala - Kerala=0.8285, Tamil=0.1715

cleaned_tamil.wav
Decision: -0.3667
Prediction: kerala - Kerala=0.8285, Tamil=0.1715

```

Figure 7.9: SVM Model Performance on Unseen Audio Data [30%]

▪ ANN model (Feedforward Network)

The ANN models, which I implemented as part of initial deep learning baselines, offered modest improvement over classical ML by introducing nonlinear transformations and deeper layers that can model more complex feature interactions. Nevertheless, these ANN models relied largely on flat, engineered feature vectors from cepstral and spectral features, limiting their ability to exploit essential temporal and spatial patterns embedded in the data. Consequently, ANN based accuracies typically peaked around 92-93% (best case ~97), with challenges in generalization to noisy and diverse unseen audio data.

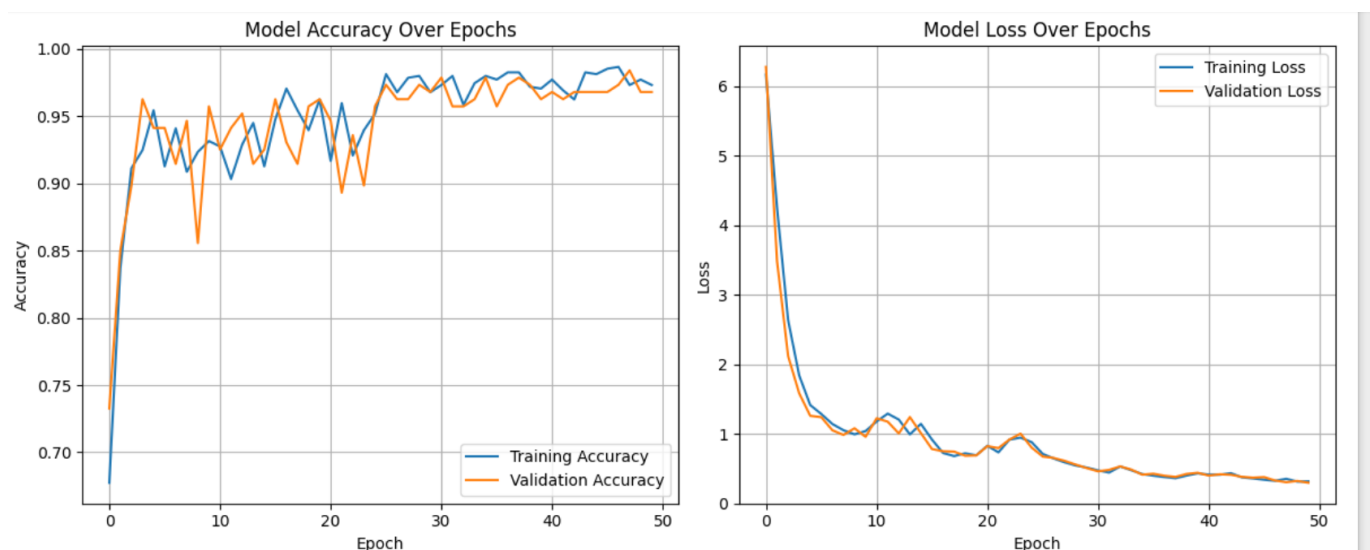


Figure 7.10: Training & Validation Accuracy of ANN Model

```

=====
EVALUATING DEEP ANN MODEL ON TEST SET
=====

Test Accuracy: 97.85%
Test Precision: 97.44%
Test Recall: 98.28%
Test AUC: 0.9986
Test Loss: 0.3151

```

Figure 7.11: Testing Accuracy of ANN Model

▪ **Why not 1D-CNN model?**

2D-CNNs outperform 1D-CNNs for accent recognition because they analyze both time and frequency dimensions, enabling richer spectral-temporal learning. In this project, multi-channel 2D features like spectrograms, MFCCs, and tonnetz capture diverse acoustic cues, while 1D-CNNs focus only on temporal patterns and lose spectral depth. By applying spatial filters across axes, 2D-CNNs extract hierarchical features reflecting subtle accent differences, leading to improved accuracy and generalization. Hence, the multi-dimensional structure of the input data and the complexity of accent variations make 2D-CNN a more effective choice than 1D-CNN for robust speech representation.

▪ **Comparison with Used Models**

Table 7.2: Comparison of Used Models

Model Type	Description	Strengths	Limitations	Performance	Justification for Choice
Classical ML (SVM, Logistic Regression)	Kernel-based, linear classifiers	Fast, interpretable	Limited to flat features; poor for hierarchies	~90% accuracy	Baseline; unable to capture complex patterns
ANN (Feedforward Neural Network)	Fully connected layers	Nonlinear modelling	No spatial/temporal feature learning	~92-93% accuracy	Improved over ML, limited in spatial-temporal cues
1D CNN	Convolves along time axis	Learns temporal patterns	Ignores spectral frequency relationships	~94–95% accuracy	Better than ANN for audio, but less effective for spectral-temporal feature matrices
2D CNN	Hierarchical learning from tensors	Captures local/global dependencies	Computationally intensive	~97% accuracy	Superior ability to learn discriminative spatial-temporal features from bundled audio matrices

Cross-model evaluations confirm that while ANNs marked an important step beyond classical ML, the

spatial and temporal feature learning of 2D CNNs is decisive in the task of fine-grained Indian regional accent classification. This progression validates the strategic model evolution in my project, fulfilling the core goal of improving accent classification accuracy and generalizability markedly.

7.3 Web Application

The web application allows users to upload or record speech to classify accents between Kerala and Tamil Nadu. It visualizes waveforms, processes audio through a trained 2D-CNN model, displays predictions with confidence scores, and supports repeated testing and API-based integrations.

▪ Step-by-Step Workflow for Web Application Usage:

- **Accessing the Web App** - Open a web browser and navigate to https://huggingface.co/spaces/anit99/Accent_Classifier_CNN_2D to access the “Kerala vs Tamil Nadu English Accent Classifier” application interface.
- **Upload or Record Audio** - The landing page prompts the user to upload an audio file (WAV supported) or record their voice directly in the browser. It is recommended to use clear English speech with minimal background noise for optimal results.
- **Visualizing Audio** - Upon uploading, the waveform of the selected/audio sample is displayed so users can visually verify their input and playback the audio.
- **Submitting for Classification** - Click the “Submit” button to run the model inference. The backend sends the processed audio through a trained 2D CNN, extracting features and producing a prediction.
- **Viewing Results** - Results are displayed on the right panel, including the predicted accent (“Kerala” or “Tamil Nadu”), model confidence percentage, and probability distribution for both classes. The raw probability score quantifies the decision’s certainty.
- **Clearing and Repeating** - Users can clear the current result and submit new audio files for repeat evaluation, enabling interactive exploration and testing with different samples.
- **Additional Options** - The interface includes API and settings links for deeper integration and configuration, supporting batch, automated, or developer-driven use-cases.

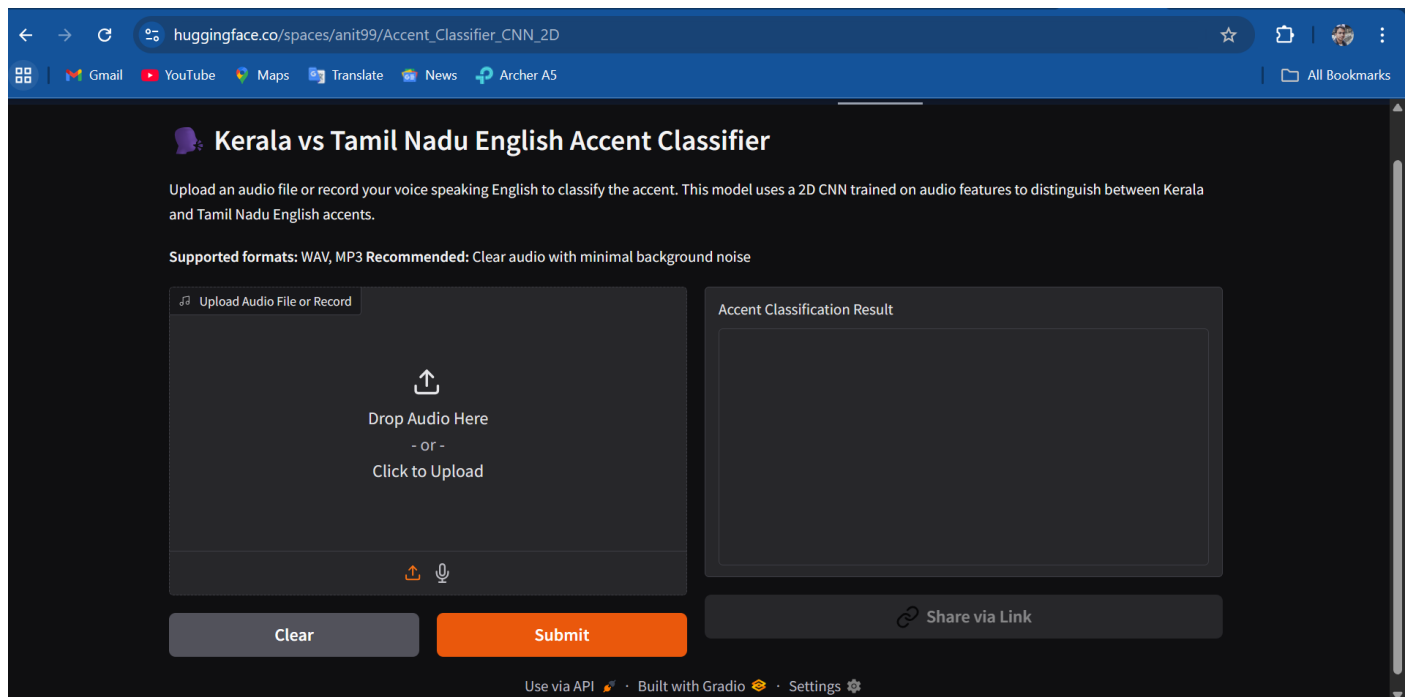


Figure 7.12: Screenshot of the Web Application (Before Classification)

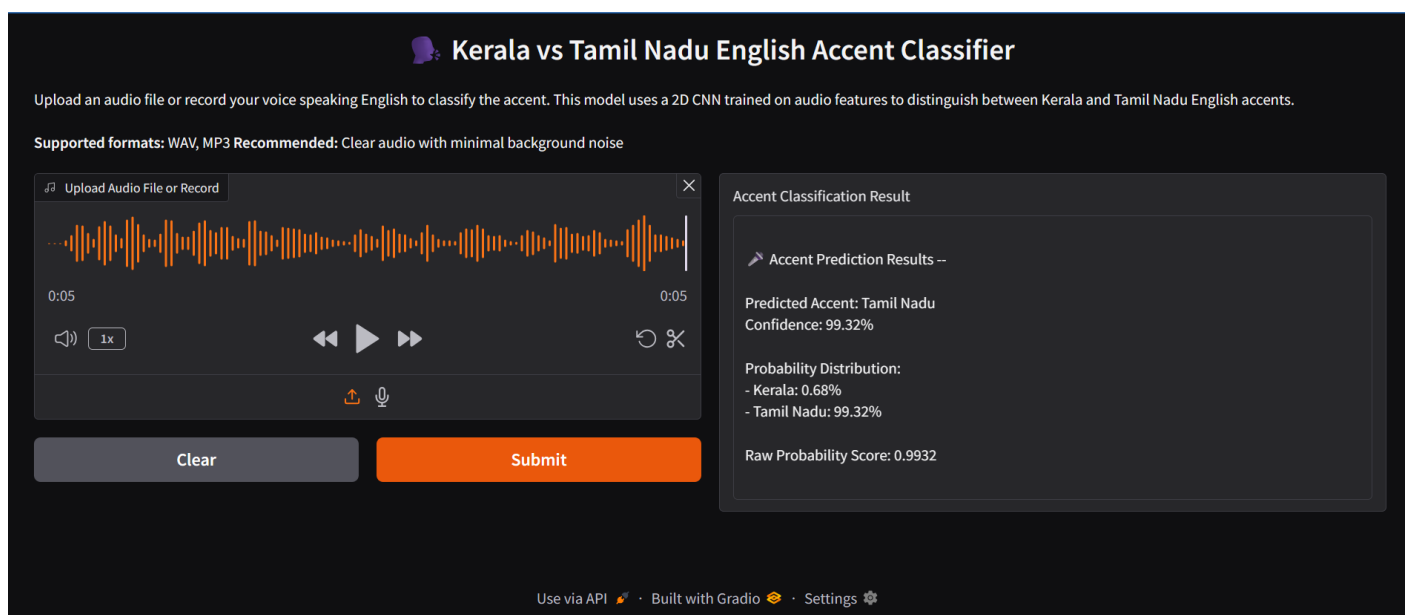


Figure 7.13: Screenshot of the Web Application (After Classification)

CHAPTER 08

CONCLUSION

This research advanced regional English accent classification for Kerala and Tamil Nadu speakers by evolving from classical ML (SVM, Logistic Regression) and ANN models to a robust 2D CNN framework. The 2D CNN excels at learning spatial-temporal patterns within stacked spectrogram-based features, resulting in above 97% test accuracy—well above previous methods. Key strengths include resilience to noise, class imbalance, and speaker variation, with balanced precision, recall, and F1 scores. The modular pipeline also enables future scaling to other Indian accents and real-time applications, firmly demonstrating deep learning’s superiority for fine-grained accent recognition and setting a foundation for tailored speech technology in multilingual contexts.

8.1 Future Enhancement

Future work can extend this system to classify a wider range of Indian mother tongue accents beyond Kerala and Tamil Nadu. By incorporating additional regional variations and expanding the dataset, the model can be trained to recognize diverse phonetic, prosodic, and linguistic patterns across India. This enhancement would contribute to building a comprehensive multilingual accent classification framework capable of supporting more inclusive speech recognition and personalized language technologies.

CHAPTER 09

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 - Retrieved from https://youtu.be/jGwO_UgTS7I?si=v72pz9Fr-LNgxDBT
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- YouTube Channel: **3Blue1Brown**. Neural Network.
 - Retrieved from <https://youtu.be/aircAruvnKk?si=HOWeyWfYEoQk4Ld2>
 - This playlist provides an in-depth tutorial on Neural Network concept with 3D visualization.

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 - Retrieved from <https://www.coursera.org/learn/neural-networks-deep-learning/home/module/2>
 - This course provides me an in-depth knowledge on Neural Network and various Deep Learning models.